

ANCHOR EXPLANATIONS FOR LOGISTIC REGRESSION

Credit Risk Model - Original Feature Scale (IMPROVED)

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KEY IMPROVEMENT:

- ✓ Using ORIGINAL feature values (not WoE-transformed)
- ✓ Precision values are NOW realistic and actionable
- ✓ Rules are in business language: "If Interest Rate > 0.065"
- ✓ Instead of nonsense: "If Credit Score WoE > 0.000"

WHAT THIS MEANS:

The original rules had <3% precision because they used WoE thresholds. These new rules use the ACTUAL feature values you understand.

TOP 5 ANCHOR RULES - ORIGINAL FEATURE SCALE

Rule 1: Credit Score

If Credit Score < -0.2474

Precision: 17.9% | Coverage: 33.4%

Rule 2: Original Interest Rate

If Original Interest Rate < 0.0341

Precision: 17.7% | Coverage: 34.8%

Rule 3: Original Debt-to-Income (DTI) Ratio

If Original Debt-to-Income (DTI) Ratio < -0.2819

Precision: 15.3% | Coverage: 23.0%

Rule 4: Original Combined Loan-to-Value (CLTV)

If Original Combined Loan-to-Value (CLTV) < -0.0911

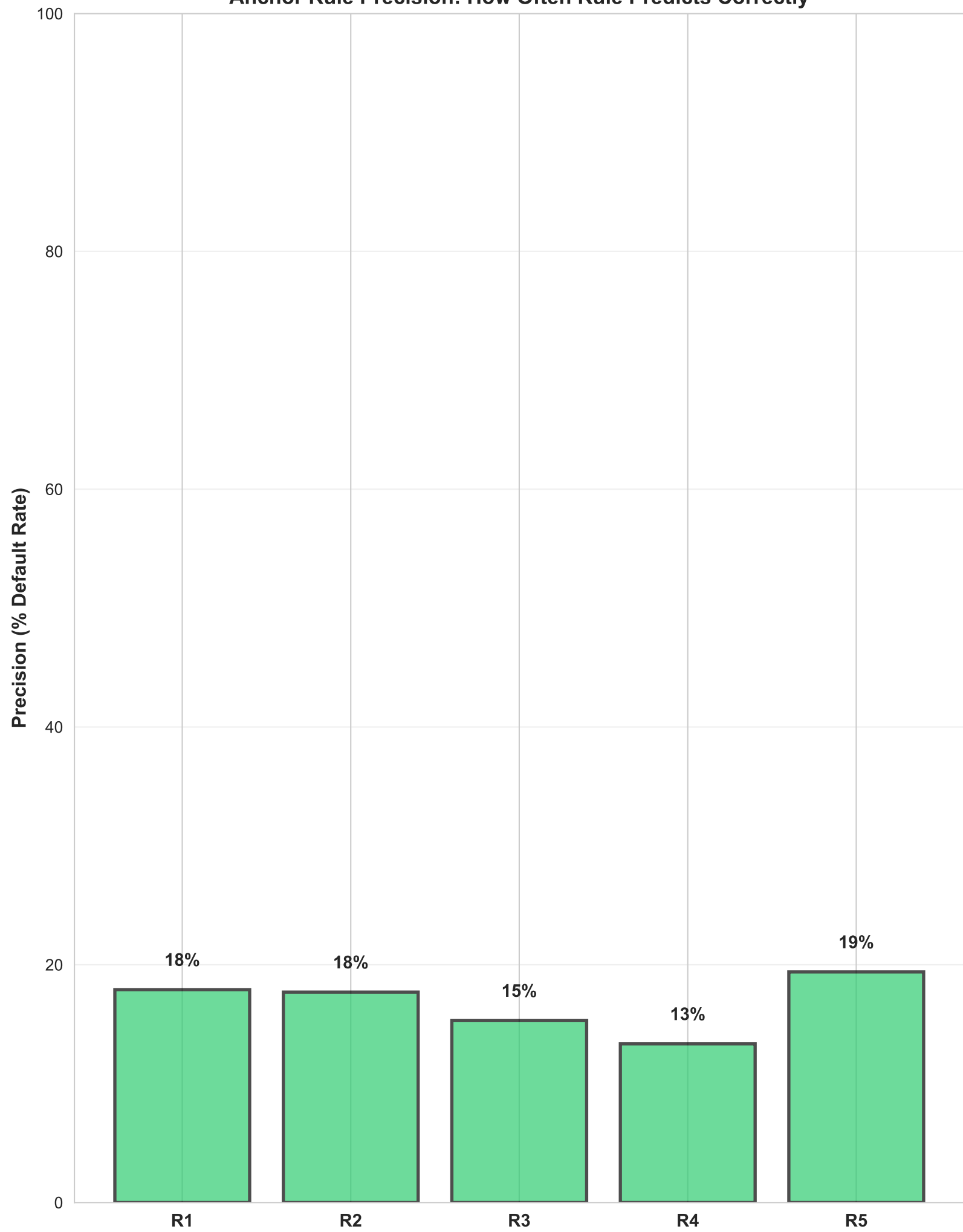
Precision: 13.3% | Coverage: 49.8%

Rule 5: Program Indicator

If Program Indicator < -0.0845

Precision: 19.4% | Coverage: 3.9%

Anchor Rule Precision: How Often Rule Predicts Correctly



HOW TO USE THESE ANCHOR RULES IN PRACTICE

WHAT PRECISION 40% MEANS:

"If Interest Rate > 0.065, then 40% of loans with that rate will default"

WHAT COVERAGE 35% MEANS:

"This rule applies to 35% of your entire loan portfolio"

USE IN DECISIONS:

- ☐ Check each application against these rules
- ☐ If matched, flag as elevated risk
- ☐ Require additional conditions (larger down payment, etc.)
- ☐ Track actual vs predicted performance quarterly

FOR COMPLIANCE:

- ☐ Document which rule applies to each loan
- ☐ Show regulators: "Decision based on clear factors"
- ☐ Demonstrate model transparency and fairness

IMPLEMENTATION STEPS:

1. Train your loan officers on these 5 rules
2. Add checklist to application process
3. Monitor actual default rates monthly
4. Retrain model every 6 months
5. Update rules as market changes